

Transfer Learning for Audio Captioning

Chen Yanxi

School of Mathematics & Physics
Qingdao University of Science
and Technology
Qingdao, China
cyx15505349500@163.com

Yang Shuguo

School of Mathematics & Physics
Qingdao University of Science
and Technology
Qingdao, China
ysg_2005@163.com

Abstract—Automatic Audio Caption Generation (AAC) aims at automatically generating linguistic descriptions for a given audio clip and is a cross-modal translation task. Most of the traditional audio caption generation methods are performed by training labeled data through supervised learning methods, but labeled data requires a lot of wasted human resources. In order to solve this problem, this paper proposes an audio caption generation method based on transfer learning training model (TLAC). The method uses the Transformer model as the encoder and decoder for the audio captioning task. First, the Log-mel spectrogram of the audio is input into the encoder, and then three masked self-supervised learning models AudioMAE, MaskSpec, and SSAST are used to train the encoder to generate effective audio features, respectively, and the audio features are input into the decoder to automatically generate the corresponding captions. In this paper, relevant experiments are conducted on AudioCaps dataset, and six evaluation metrics are used to evaluate the effect of caption generation, and the results show that the method in this paper can reach the performance of supervised learning methods, and the accuracy and smoothness of the generated captions are greatly improved.

Keywords—Automatic audio captioning, transfer learning, cross-modal tasks, self-supervised learning, natural language processing

I. INTRODUCTION

Automatic Audio Captioning (AAC) generation is not only used to portray the content of audio clips, but also to reflect the temporal sequence and the interaction of audio scenes [1]. In recent years, this task has been widely used in real-life applications and attracted more and more attention. Currently, audio caption generation faces a dual challenge: on the one hand, the audio dataset with labeled captions is small in size, much smaller than those in the fields of computer vision or natural language processing, which restricts the performance of caption generation; on the other hand, traditional supervised learning methods rely on a large number of manual annotations, which leads to too large a training cost and poor results. Transfer learning can solve this problem by taking advantage of data from other related tasks. Therefore, in this paper, we adopt Self-Supervised Learning (SSL) in caption generation, which not

only breaks through the strong dependence of traditional supervised learning on labeled data, but also improves the model's understanding of audio semantics, and provides a new way of thinking for audio caption generation.

Currently, a variety of SSL models have been proposed. He et al [2] proposed the AudioMAE method in audio self-supervised learning, which achieved good results on various tasks in audio; Chong et al [3], also inspired by MAE, proposed a pre-training target MaskSpec that directly recovers the masked blocks of the spectrogram, which demonstrated good performance in the downstream tasks. Later, Gong et al [4] proposed the SSAST structure and introduced the patch-based (Patch) and fully self-attentive mechanisms to the audio domain for the first time. In this paper, three typical SSL models AudioMAE, MaskSpec, and SSAST are applied to the audio captioning task, and the TLAC_A, TLAC_M, and TLAC_S models are constructed respectively.

In this paper, the TLAC model is experimented on AudioCaps dataset and compared with typical supervised pre-training models ACT_Deit_AudioSet [5], CNN+Transformer [6], RNN+RNN [7], CNN+RNN [8], in comparison, the AudioMAE model in this paper generates the best results, and its BLEU₁ index is better than the ACT_Deit_AudioSet model, and other indexes are similar to the ACT_Deit_AudioSet model, reflecting the effectiveness of the method.

II. TRANSFERRING SELF-SUPERVISED LEARNING TRAINING METHODS

In this paper, the Transformer model architecture is used. The original audio is transformed to get its Log-Mel spectrogram, the spectrogram is reconstructed, and the Vision Transformer (ViT) encoder is used to encode it to output the audio features, which are then fed into the lightweight Transformer decoder, meanwhile, the original captions are added with the positional information and then fed into the decoder as well, and the two outputs go through the multi-attention mechanism, residual and normalization operations to generate captions. The specific flow is shown in Figure 1.

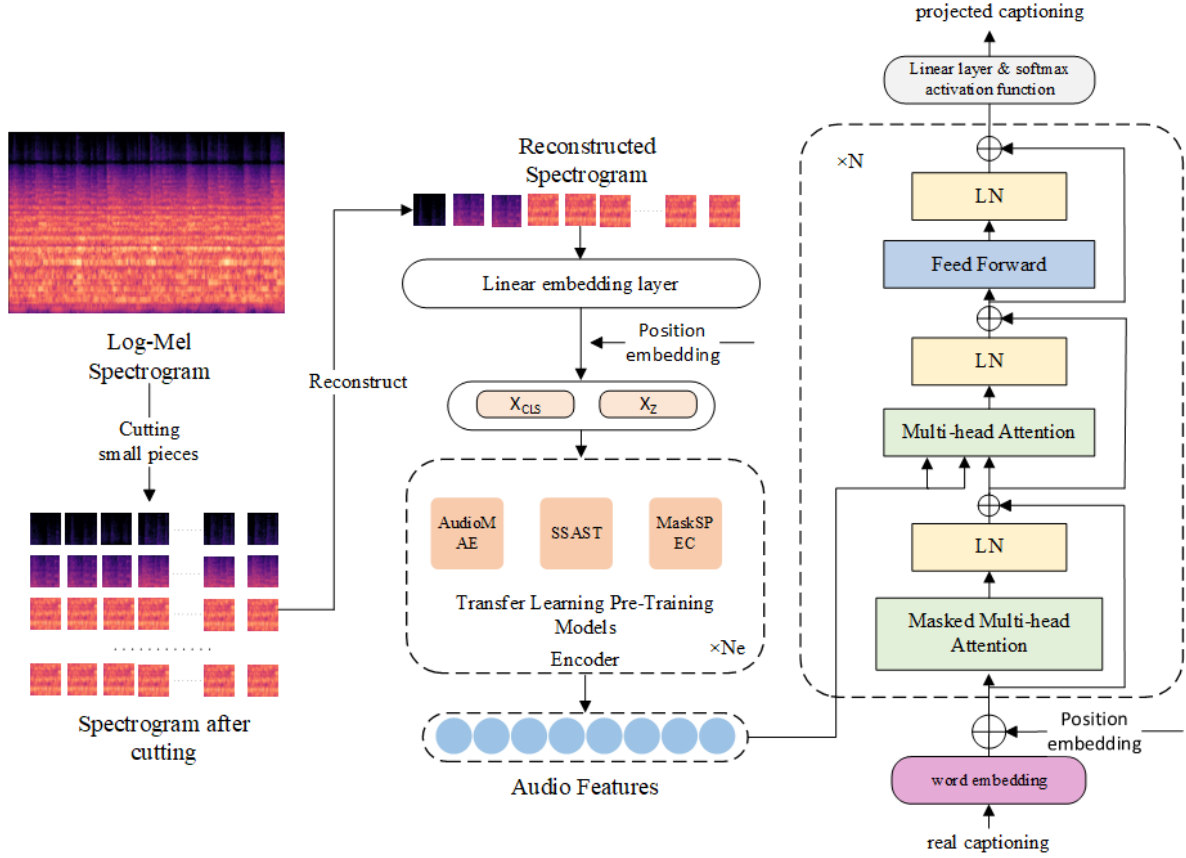


Fig. 1. Overall framework of the TLAC model

A. Encoder

The original audio is preprocessed and converted into a Log-Mel spectrogram $X \in \mathbb{R}^{T \times F}$, where T is the number of time frames and F is the Mel filter. The Log-Mel spectrogram is able to extract the key frequency information in the audio signal and present it visually, which is easy to analyze and understand. To adapt the sequence processing mechanism of the Transformer architecture, a sliding window operation [9] is used to partition X into Patch $X_i (i = 1, 2, \dots, N)$, $N = \lceil T/P \rceil \times \lceil T/P \rceil$ of size $P \times P$. Each Patch is converted to an P^2 -dimensional vector by the spreading operation of the linear embedding layer to construct the chunked feature matrix $X' \in \mathbb{R}^{N \times P^2}$. Drawing on the embedding method of ViT, a learnable linear projection $W_E \in \mathbb{R}^{P^2 \times d}$ is used to map each Patch to the hidden space, generating the serialized embedding representation $X_Z = X'W_E \in \mathbb{R}^{N \times d}$, where d is the model hidden layer dimension. Add learnable global vectors $X_{cls} \in \mathbb{R}^{1 \times d}$, and use fixed sinusoidal positions $X_{pos} \in \mathbb{R}^{(N+1) \times d}$ to label the position information of X_Z at the forefront of the spread one-dimensional vectors X_Z , thus obtaining the input vector representation as:

$$X_e = [X_{cls} + X_Z] + X_{pos} \in \mathbb{R}^{(N+1) \times d}. \quad (1)$$

The vector X_e is fed into the ViT encoder, and the encoder is pretrained by the self-supervised learning models AudioMAE, MaskSpec, and SSAST, respectively. The encoder core contains $N_e = 4$ coding blocks, each of which consists of two residual and normalization layers, a multi-head attention mechanism layer, and a feed-forward network. The most important part of the coding block is the multi-head self-attention mechanism layer, which consists of multiple self-attention mechanisms. Then the three learnable matrices W of the multi-head self-attention mechanism layer are applied to X to generate three vectors of Q -query, K -key, and V -value, which are computed as follows:

$$\begin{cases} Q = W^Q X_e, \\ K = W^K X_e, \\ V = W^V X_e. \end{cases} \quad (2)$$

After obtaining the Q -query, K -key, and V -value, the output from the self-attention mechanism layer can be computed

$$Attn(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \quad (3)$$

where $\text{softmax}(\cdot)$ denotes the activation function and d_k denotes the dimension of the Q -query, K -key, and V -value.

In Equation (3), the query vector Q is inner product with the key vector K . To prevent the inner product from being too large divide it by the square root of d_k . The matrix can represent the attention intensity between Patches. This matrix can represent the intensity of attention between Patches; finally the softmax activation function is used to calculate the attention coefficients between each Patch and each Patch, and the sum of each row of the matrix is 1. Multiplying the attention coefficients with the value vectors V yields the output of the Self-attention mechanism layer. While the multi-head attention mechanism is to splice the outputs of multiple self-attention mechanism layers and finally pass into a linear layer, which finally gives the output of the multi-head attention mechanism. This output matrix can be expressed as:

$$MH(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W_o, \quad (4)$$

$$\text{head}_j = \text{Attn}(QW_j^Q, KW_j^K, VW_j^V). \quad (5)$$

where $j = 1, 2, \dots, h$, we set $h = 12$ in this paper.

The FeedForward network in the ViT encoder consists of two linear transformations with GELU activation functions. A residual connection and a normalization layer are connected before and after the self-attention mechanism layer and the feedforward network module. Since the ViT contains a multi-layer structure, the residual connection can focus on the difference part of the current layer and reduce the computational complexity. Normalization, on the other hand, accelerates the convergence of the model by normalizing the inputs of each layer to have the same mean and variance. The combination of these two not only enhances the computational efficiency of the model, but also improves the stability of the training process, and the final output is:

$$X_{out} = \text{LayerNorm}(X_{in} + f(X_{in})). \quad (6)$$

where $f(\cdot)$ denotes the output of the multi-head attention mechanism and the feed-forward network, X_{in} denotes the input of each layer, and X_{out} denotes the residual connection and the output of the normalization layer.

The final output yields the audio feature extracted by the encoder, which captures the long-term dependencies in the sequence data and facilitates the decoder to generate high-quality audio captions.

B. Self-supervising Model

In order to obtain a more detailed representation of audio features, this paper performs transfer learning during encoder

training. The existing AudioMAE, MaskSpec, and SSAST self-supervision are utilized for transfer. the AudioMAE model adopts a high proportion of random block masking to enhance global structural inference; the MaskSpec model enhances local feature sensitivity through time-frequency 2D convolutional masking; and the SSAST model introduces a semantically-guided masking mechanism to improve the sound content recognition capability. Since the principles and architectures of the three methods are basically similar, the AudioMAE model, for example, is introduced as follows.

The method employs a spectrogram-based masking strategy $m(\cdot)$ to partially mask the input audio, generate a masked spectrogram and input it to the encoder $Enc(\cdot)$. The design significantly reduces the complexity and convergence difficulty of the encoder computation for large-scale datasets by reducing the effective input length of the model. The potential audio features extracted by the encoder are combined with the positional coding information and fed into the decoder in conjunction with the masked markers and the masked spectrogram is reconstructed in the decoding stage. The goal of the method is to minimize the loss function between the predicted spectrogram r_i and the input spectrogram x_i , which can be expressed as:

$$\text{Loss} = \frac{1}{N} \sum_{i=1}^N (r_i - x_i)^2. \quad (7)$$

In this paper, three self-supervised learning models are used to pre-train the encoder through a parameter transfer strategy [10-11], and the trained audio features are utilized to improve the accuracy and smoothness of audio captioning. In this way, we can get similar results to the supervised pre-training models while reducing the use of labeled data and the waste of manual resources for caption annotation.

C. Decoder

The decoding part of the TLAC model in this paper consists of a word embedding layer, a decoder, and a linear layer. The word embedding layer is a pre-trained Word2vec model, and the captions are transformed into word vectors through the word embedding layer, which provides rich semantic feature representations for the captions words. The decoder in this paper uses an $N = 4$ layer Transformer decoding block to generate the corresponding linguistic descriptions for the audio clips output from the encoder. The output of the decoding block passes through a linear layer and subsequently the similarity between individual words is computed using a softmax activation function, and finally the model takes the word with the highest similarity as the output, which is obtained from the final linear layer. The goal of the model training is to optimize the parameters by minimizing the cross-entropy loss function to achieve highly accurate prediction of the real caption y . Such a training process aims to improve the predictive ability of the model so that it can more accurately reflect the semantic information of the audio content when automatically generating captions. The formula for the cross-entropy loss function is as follows:

$$L_{CE}(\theta) = -\frac{1}{T} \sum_{i=1}^T \log p(y_i | y_{1:i-1}, \theta). \quad (8)$$

III. RELATED EXPERIMENT

A. Dataset

The AudioSet dataset is a large-scale audio event recognition dataset developed by a team of Google researchers, which has an ontology of 527 sound classes and contains more than 2 million 10-second audio clips extracted from YouTube videos.

AudioCaps [12] has about 50k audio clips from AudioSet with manually labeled captions. It contains three parts: training set, test set and validation set. Each audio clip in the training set contains one manual caption, and the audio clips in the test and validation sets contain five captions. The TLAC models in this paper are experimented with this dataset.

B. Data Pre-processing

All audio clips in the dataset were converted to 16k Hz and padded to 10 seconds long. The Log-Mel spectrogram is then computed using a 1024-point Hanning window with 50% overlap and 64 Mel bins, leading to input features of shape 1024×64 . Each Log-Mel spectrogram is split into 128 patches along the time axis and 4 patches along the frequency axis, for a total of 128×4 patches with a patch size of 16×16 . The obtained two-dimensional matrix (128×4 , 16×16) is linearly transformed with the learnable matrix (16×16 , 768) to obtain a one-dimensional embedding (128×4 , 768). Input features are augmented during training using SpecAugment [13]. Captions are tokenized and converted to lowercase, with punctuation removed. To indicate the beginning and end of each caption, two special tags “<eos>” and “<eos>” are padded. The AudioCaps vocabulary contains 5,277 different words.

C. Experimental Setup

In this paper, the Adam optimizer is used to train the model for 60 rounds, with the batch size set to 32. The initial learning rate is set separately for different self-supervised learning models to fit the parameter updates; the robustness of the optimization process is improved by using a 5-round linear learning rate preheating strategy, and then after 30 rounds, the model enters into an exponential decay phase, where the learning rate is scaled down to 10% of the original value every 10 rounds. To prevent model overfitting, a Dropout probability of 0.2 is configured in each feedforward network layer of the encoder-decoder, and a label smoothing technique with a smoothing coefficient of 0.1 is also used to prevent model overfitting. In the inference stage, the accuracy of generated captions is optimized by dynamically adjusting the beam search strategy (maximum beamwidth of 5).

D. Evaluation Indicators

In this paper, the evaluation of TLAC model adopts the evaluation metrics related to image captioning task, including the machine translation metrics BLEU_n, ROUGE_L, METEOR and the specialized captioning metrics CIDE_r, SPICE, SPIDE_r. BLEU_n is based on the type of N-gram, and introduces a penalty factor to control the score of phrases; ROUGE_L calculates F-value through the longest common subsequence; METEOR calculates the average of precision and recall, and supports synonym matching; CIDE_r is the TF-IDF weighted N-gram vector cosine similarity, and evaluates precision and recall, and focuses on recall evaluation; METEOR calculates the reconciled average of precision and recall to support synonym matching; CIDE_r is the TF-IDF weighted N-gram vector cosine similarity to evaluate semantic relevance; SPICE calculates F-values based on scene graph tuples to verify semantic fidelity; and SPIDE_r is the weighted average of CIDE_r and SPICE, which takes into account both grammatical fluency with semantic accuracy.

IV. ANALYSIS OF EXPERIMENTAL RESULTS

In this paper, TLAC models are evaluated on the AudioCaps dataset. The impact of three different self-supervised learning models on the performance of caption generation is analyzed and compared with supervised pre-training models, and the experimental results are shown in Table I.

A. Comparison between the Performance of Different Self-supervised Training Models

In this paper, three self-supervised pre-training models AudioMAE, MaskSpec, and SSAST are used to pre-train the audio caption generation model. The experimental results in Table 1 show that the optimal performance is achieved with the TLAC_A model, and its advantages mainly come from two aspects: (1) the model enables the encoder to reconstruct the complete Log-Mel spectrogram from the discrete acoustic coding through the 75% high percentage random block masking strategy, which significantly strengthens the global modeling capability; (2) the generalized ViT-based architectural design enables the model to efficiently capture the cross-modal fine-grained audio features required for alignment, thus enhancing the semantic coherence of downstream caption generation.

In addition, the TLAC_M model uses a time-frequency two-dimensional adaptive masking with contrast learning strategy, which is advantageous in modeling local acoustic patterns, while the TLAC_S model locates the content of key audio clips through a dynamic masking mechanism. In this paper, experiments with the TLAC_scratch model without self-supervised model training are also conducted (results are shown in row 4 of Table I). The results show that all the metrics of the three self-supervised pretraining models have significant performance improvements over those of TLAC_scratch, proving the feasibility and effectiveness of self-supervised pretraining in audio captioning tasks.

TABLE I. COMPARISON RESULTS OF TLAC MODELS AND SUPERVISED MODELS

	BLEU ₁	BLEU ₂	BLEU ₃	BLEU ₄	ROUGE _L	METERO	CIDE _r	SPICE	SPIDE _r
TLAC_A	0.654	0.483	0.348	0.240	0.465	0.219	0.631	0.160	0.396

TLAC_M	0.647	0.474	0.340	0.239	0.470	0.239	0.625	0.161	0.393
TLAC_SS	0.651	0.479	0.337	0.237	0.465	0.220	0.607	0.159	0.383
TLAC_scratch	0.572	0.391	0.262	0.173	0.406	0.178	0.424	0.123	0.274
ACT_Deit_AudioSet	0.653	0.495	0.363	0.259	0.471	0.222	0.663	0.163	0.413
CNN+RNN	0.655	0.476	0.335	0.231	0.467	0.229	0.660	0.168	0.414
CNN+Transformer	0.641	0.479	0.344	0.236	0.469	0.221	0.693	0.159	0.426
CNN+Transformer	0.641	0.479	0.344	0.236	0.469	0.221	0.693	0.159	0.426

B. Comparison with Supervised Training Models

In this paper, the TLAC method is compared with the representative supervised training methods ACT_Deit_AudioSet and CNN+Transformer. First, by comparing with the structurally similar supervised pre-training model ACT_Deit_AudioSet, the results show that under the similar codec architecture, the TLAC with transferred self-supervised pre-training only has a small difference within 3% with the ACT_Deit_AudioSet model in various indexes, which verifies that self-supervised pre-training method can effectively replace the traditional supervised learning, and at the same time demonstrates the model's ability to generalize and transfer cross-task knowledge.

In addition, TLAC compares with previous representative model structures for audio annotation, RNN + RNN, CNN + RNN, and CNN + Transformer, and demonstrates its advantages. Compared to RNN + RNN, the results of TLAC for all metrics are superior to its model. For BLUE_{2,3,4} performance metrics,

TLAC is higher than the CNN + RNN model, indicating that TLAC can generate smoother annotations. Compared to the more advanced CNN + Transformer model, TLAC similarly improves the model's ability in machine translation. Since machine translation metrics are mainly based on n-grams and the self-supervised model learns features by reconstructing the generated spectrograms, it gives TLAC the ability to better generate accurate words.

This paper also explores the effect of changing the number of decoder layers N and the number of heads h of the multi-head attention mechanism on the model results for analysis. Three sizes of decoders are selected in the paper, De_s (2 layers and 4 heads), De_m (4 layers and 8 heads), and De_l (6 layers and 8 heads). The results in Table II show that the decoder structure size of De_m outperforms the other two decoder sizes in all evaluation metrics, because the amount of data in audio captions is small and each annotation is mostly a short sentence, so too large or too small decoders may affect the caption prediction results.

TABLE II. EFFECT OF DECODER SIZE ON TLAC_AUDIOMAE MODEL RESULTS

Decoder	BLEU ₁	BLEU ₂	BLEU ₃	BLEU ₄	ROUGE _L	METERO	CIDE _r	SPICE	SPIDE _r
De-S	0.647	0.466	0.329	0.227	0.460	0.216	0.608	0.160	0.384
De-B	0.654	0.483	0.348	0.240	0.465	0.219	0.631	0.160	0.396
De-L	0.634	0.458	0.322	0.221	0.455	0.214	0.579	0.149	0.363

V. CONCLUSION

In this paper, we propose an audio captioning model (TLAC) based on transferred learning model, whose main advantage lies in the utilization of unlabeled data for pre-training, thus significantly reducing the consumption of manual captioning resources on which the traditional supervised learning model relies. The experimental results show that the strategy of using transferred self-supervised learning models can effectively improve the model performance when compared with the TLAC_scratch model, demonstrating the potential of self-supervised learning models for audio generated captioning tasks and their competitiveness with existing state-of-the-art models. However, compared to the latest supervised training methods, TLAC still suffers from a 10% CIDE_r performance gap in complex acoustic scenarios. This is due to the bias in the alignment of representations between the self-supervised learning target and the downstream generation task, illustrating

that self-supervised learning still has a large room for improvement on the audio captioning task.

REFERENCES

- [1] Mei X, Liu X, Plumbley M D, et al. Automated audio captioning: An overview of recent progress and new challenges[J]. EURASIP journal on audio, speech, and music processing, 2022, 2022(1): 26.
- [2] Huang P Y, Xu H, Li J, et al. Masked autoencoders that listen[J]. Advances in Neural Information Processing Systems, 2022, 35: 28708-28720.
- [3] Chong D, Wang H, Zhou P, et al. Masked spectrogram prediction for self-supervised audio pre-training[C]//ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2023: 1-5.
- [4] Gong Y, Lai C I, Chung Y A, et al. Ssast: Self-supervised audio spectrogram transformer[C]//Proceedings of the AAAI Conference on Artificial Intelligence. 2022, 36(10): 10699-10709.
- [5] Mei X, Liu X, Huang Q, et al. Audio captioning transformer[J]. arXiv preprint arXiv:2107.09817, 2021.
- [6] Chen K, Wu Y, Wang Z, et al. Audio Captioning Based on Transformer and Pre-Trained CNN[C]//DCASE. 2020: 21-25.

- [7] Drossos K, Adavanne S, Virtanen T. Automated audio captioning with recurrent neural networks[C]//2017 IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA). IEEE, 2017: 374-378.
- [8] Mei X, Huang Q, Liu X, et al. An encoder-decoder based audio captioning system with transfer and reinforcement learning[J]. arXiv preprint arXiv:2108.02752, 2021.
- [9] Dosovitskiy A, Beyer L, Kolesnikov A, et al. An image is worth 16x16 words: Transformers for image recognition at scale[J]. arXiv preprint arXiv:2010.11929, 2020.
- [10] Xu X, Dinkel H, Wu M, et al. Investigating local and global information for automated audio captioning with transfer learning[C]//ICASSP 2021-2021 IEEE international conference on acoustics, speech and signal processing (ICASSP). IEEE, 2021: 905-909.
- [11] Won H, Kim B, Kwak I Y, et al. Transfer learning followed by Transformer for automated audio captioning[C]//Proceedings of the 6th Detection and Classification of Acoustic Scenes and Events 2021 Workshop (DCASE2021), Barcelona, Spain. 2021: 221-225.
- [12] Kim C D, Kim B, Lee H, et al. AudioCaps: Generating captions for audios in the wild[C]//Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers). 2019: 119-132.
- [13] Park D S, Chan W, Zhang Y, et al. SpecAugment: A simple data augmentation method for automatic speech recognition[J]. arXiv preprint arXiv:1904.08779, 2019.